26 Sean: The Thom Isomorphism Theorem

26.1 What happens when M is not compact?

Let us recall the statement of Poincare duality from class:

Theorem 54. Let M^n be a closed, orientable manifold. Then there is a non-degenerate (perfect) pairing $\langle \cdot, \cdot \rangle : H^k_{dR}(M) \otimes H^{n-k}_{dR}(M) \to \mathbb{R}$ given by

$$\langle \omega, \eta \rangle = \int_M \omega \wedge \eta.$$

What did this mean in particular? Let $\langle \cdot, \cdot \rangle : V \otimes W \to \mathbb{R}$ be a pairing of finite dimensional vector spaces. It is nondegenerate if $\langle v, w \rangle = 0$ for all $w \in W$ implies that $v = 0$ and likewise, $\langle v, w \rangle = 0$ for all $v \in V$ implies $w = 0$. In other words,

$$v \mapsto \langle v, \cdot \rangle, \quad w \mapsto \langle \cdot, w \rangle$$

define injections from $V \hookrightarrow W^*$ and $W \hookrightarrow V^*$. This turns out to be equivalent to saying that the map $v \mapsto \langle v, \cdot \rangle$ defines an isomorphism from V to W^*. So in this point of view, we get the isomorphism

$$H^k_{dR}(M) \cong (H^{n-k}_{dR}(M))^*.$$

What happens when M is not compact (i.e. it is not closed)? The first thing that breaks is integration, since we have only defined integration for compactly supported top forms on smooth, oriented manifolds.

To handle this non-compact case, we need a new cohomology theory. It looks like de Rham cohomology, but with a few key differences. Indeed, let us define the vector subspace

$$\Omega^k_c(M) = \{\text{compactly supported smooth } k\text{-forms} \in \Omega^k(M)\},$$

and the cohomology groups

$$H^k_c(M) = \frac{\ker(d : \Omega^k_c(M) \to \Omega^{k+1}_c(M))}{\operatorname{im}(d : \Omega^{k-1}_c(M) \to \Omega^k_c(M))}.$$

Obviously, when M is a compact manifold, the two cohomology theories coincide.

Question: Are these two cohomology theories really different? At first glance, seems like one should be a sub-case or generalization of the other.

Answer: Yes. Notice that since $\Omega^k_c(M) \subset \Omega^k(M)$, we get a natural map $i : H^k_c(M) \to H^k_{dR}(M)$ when we descend to cohomology. However, i need not be injective or surjective necessarily.

We now summarize the properties of compactly supported cohomology in a table:

Property	de Rham cohomology	Compactly supported cohomology
Functorality	contravariant	covariant
Homotopy invariance	Yes	No
Diffeomorphism invariance	Yes	Yes
Poincare lemma	$H^k(M \times \mathbb{R}^n) = H^k(M)$	$H^k_c(M \times \mathbb{R}^n) = H^{k-n}_c(M)$

26.2 The new Mayer-Vietoris sequence

Note that if $f \in C^\infty(M)$ and ω is some form with compact support on M, then $f^*\omega$ need not have compact support in M. Hence, we cannot "functorally" send smooth, compactly supported maps between manifolds to their pullbacks like we do for normal de Rham cohomology.

Example 50. Consider the projection $\pi : M \times \mathbb{R} \to M$ and some smooth, compactly supported function $f \in C^\infty_c(M)$ (a 0-form). Then,

$$\pi^* f(x,t) = f(x)$$

for all $(x,t) \in M \times \mathbb{R}$, so we see that

$$\operatorname{supp}(\pi^* f) = \operatorname{supp}(f) \times \mathbb{R},$$

which is not compact.

What is the solution? Instead of allowing all $f \in C^\infty_c(M)$, we restrict our attention to *inclusions of open sets* $j : U \hookrightarrow M$. In particular, we define the map $j_* : \Omega^k_c(U) \to \Omega^k_c(M)$ by extending the form by zero outside of U:

$$(j_*\omega)(p) = \begin{cases} \omega(p) & p \in U, \\ 0 & p \notin U. \end{cases}$$

While the Ω^* functor is contravariant (and hence H^*_{dR} too), the Ω^*_c functor is *covariant*. In other words, all the arrows in the Mayer-Vietoris sequence will swap. Indeed, consider the following diagrams:

$$\begin{tikzcd} & U \arrow[dr, hook, "k"] & \\ U \cap V \arrow[ur, hook, "i"] \arrow[dr, hook, "j"'] & & U \cup V = M \\ & V \arrow[ur, hook, "l"'] & \end{tikzcd} \qquad \begin{tikzcd} & \Omega^*_c(U) \arrow[dr, hook, "k_*"] & \\ \Omega^*_c(U \cap V) \arrow[ur, hook, "i_*"] \arrow[dr, hook, "j_*"'] & & \Omega^*_c(M) \\ & \Omega^*_c(V) \arrow[ur, hook, "l_*"'] & \end{tikzcd}$$

Then, our Mayer-Vietoris sequence becomes:

$$
\begin{tikzcd}
H_c^{k+1}(U \cap V) \arrow[r, "-j_* \oplus i_*"] & H_c^{k+1}(U) \oplus H_c^{k+1}(V) \arrow[r, "k_*\alpha_1 + l_*\alpha_2"] & \cdots \\
H_c^{k}(U \cap V) \arrow[r, "-j_* \oplus i_*"] & H_c^{k}(U) \oplus H_c^{k}(V) \arrow[r, "k_*\alpha_1 + l_*\alpha_2"] & H_c^{k}(M) \arrow[ull, rounded corners, "d_*"] \\
\cdots \arrow[r, "-j_* \oplus i_*"] & H_c^{k-1}(U) \oplus H_c^{k-1}(V) \arrow[r, "\cdots"] & H_c^{k-1}(M) \arrow[ull, rounded corners, "d_*"]
\end{tikzcd}
$$

26.3 Poincare duality revisited

Now, we can restate Poincare duality as follows:

Theorem 55. Let M be an orientable manifold. Then there is a non-degenerate pairing $\langle \cdot, \cdot \rangle : H^k_{dR}(M) \otimes H_c^{n-k}(M) \to \mathbb{R}$ given by

$$\langle \omega, \eta \rangle = \int_M \omega \wedge \eta.$$

In particular, we have

$$H^k(M) \cong (H_c^{n-k}(M))^*.$$

26.4 Cohomology of vector bundles and the Thom class

Recall that we can always deformation retract a rank n vector bundle $\xi : E \xrightarrow{\pi} M$ along the zero-section. By the homotopy invariance of de Rham cohomology, this means that the cohomology of any vector bundle over a manifold M is equivalent to the cohomology of M. Hence, if we want to develop any interesting theory of the cohomology of vector bundles, we must work with compactly supported cohomology.

Indeed, let $\xi : E \xrightarrow{\pi} M^m$ be an oriented rank n vector bundle over a compact, oriented manifold M.

Definition 91. A *Thom class* for ξ is a cohomology class $\Phi \in H_c^n(E)$ such that for every $p \in M$, we have that

$$\int_{E_p} \Phi|_{E_p} = \int_{E_p} i_p^* \Phi = 1,$$

where $i_p : E_p \hookrightarrow E$ is the inclusion of the fiber.

Intuitively, a Thom class for a vector bundle is a kind of "delta form" that integrates to 1 on each fiber and is supported near the 0-section.

Proposition 43. The *Thom class* is the unique class satisfying its defining property.

26.5 The Thom isomorphism

Recall that $\pi : E \to M$ induces the map $\pi^* : H^k(M) \to H^k(E)$, which turns out to be an isomorphism due to the fact that E deformation retracts onto M.

Theorem 56 (Thom isomorphism)**.** If $\xi : E \xrightarrow{\pi} M$ is an oriented rank n vector bundle over an oriented manifold M, then the map

$$H^k(M) \to H_c^{k+n}(E), \quad [\omega] \mapsto \pi^*[\omega] \smile \Phi$$

is an isomorphism for all k.

Corollary 20. In particular, for trivial bundles, we have:

$$H^k(M) \cong H_c^{k+n}(M \times \mathbb{R}^n).$$

Example 51. This works best when our bundle is nontrivial. For instance, take $TS^2 \to S^2$. Since TS^2 is an oriented rank-2 vector bundle, the Thom isomorphism gives

$$H^k(S^2) \cong H_c^{k+2}(TS^2),$$

but because $H^0(S^2) \cong \mathbb{R}$ and $H^2(S^2) \cong \mathbb{R}$, we get that

$$H_c^2(TS^2) \cong \mathbb{R}, \quad H_c^4(TS^2) \cong \mathbb{R},$$

and all the other compactly supported cohomology groups vanish. Using the Euler class, you can even prove that TS^2 is nontrivial, even if you did not know this a priori.

Now comes the big question: what exactly is Φ? Recall from Aidan's talk that if L is a embedded closed submanifold of dimension k in M, then we have a map

$$\int_L : \Omega^k(M) \to \mathbb{R}$$

which descends to cohomology. In particular, by Poincare duality, there is a form $\eta_L \in H^{n-k}(M)$ such that

$$\int_L \omega = \int_M \omega \wedge \eta_L$$

for all closed forms $\omega \in \Omega^{n-k}(M)$, and we write $\eta_L = [L]$ (the cycle class map). We call this η_L the Poincare dual of L.

Proposition 44. The Poincare dual of $L^k \subset M^n$ is the Thom class of the normal bundle of L. Indeed, $NL \to L$ is a vector bundle of rank $n-k$, and if $\Phi \in H_c^{n-k}(NL)$ is its Thom class, then

$$\eta_L = j_*\Phi \in H^{n-k}(M),$$

where $j_*\Phi$ denotes the extension of Φ by 0 outside a tubular neighborhood of NL.

Notice that the Thom isomorphism gives us the Thom class by

$$H^0(L) \to H_c^{n-k}(NL), \quad [1] \mapsto [1] \smile \Phi = \Phi.$$

Proposition 45. The Poincare dual of a closed oriented submanifold S in an oriented manifold M and the Thom class of the normal bundle of S can be represented by the same forms.

Proposition 46 ("Main theorem of intersection theory")**.** If R and S are closed, oriented submanifolds of M that intersect transversally, then

$$\eta_{R\cap S} = \eta_R \wedge \eta_S.$$

Proof. We have

$$N(R \cap S) = NR \oplus NS,$$

so we have the Thom classes:

$$\Phi(N(R \cap S)) = \Phi(NR \oplus NS) = \Phi(NR) \wedge \Phi(NS),$$

so by the previous proposition, we get that

$$\eta_{R\cap S} = \eta_R \wedge \eta_S.$$

□